

MTH310/MTH520: Graph Theory (GT)

Quiz 4

April 11, 2025

Solve either one of Q1 or Q2 in your answer booklet. In case both are attempted, the answer with the maximum marks will be taken as the final score.

Question 1.1. [CO2,CO3] Prove or disprove the following questions:

- a) There exists a tree with 2 possible perfect matchings (1-factors).

(Hint: Assume there are two possible perfect matchings M and M' . Consider the graph obtained by keeping all the vertices of the tree, but only the edges in $M \Delta M'$. Noting that both M and M' are perfect matchings, what can you say about the degree of the vertices in this new graph? Can you see a contradiction ahead?)

- b) $\forall n \in \mathbb{N}, \chi(Q_n) = n$

- c) For every bipartite graph G , there exists a matching of size at least $\frac{|E(G)|}{\Delta(G)}$.

- d) Every bipartite graph has a unique bipartition (A, B) (apart from (B, A)).

- e) $\forall k \in \mathbb{N}$, there exists a connected graph G such that $\tau(G) = k$.

[2+2+2+2+2 = 10 points]

OR

Question 1.2. [CO1,CO3] For any arbitrary graph G , prove that the following hold:

- a) $\chi'(G \square K_2) = \Delta(G \square K_2)$

- b) $\chi'(G \square H) = \Delta(G \square H)$, where H is any Class 1 graph ($\chi'(H) = \Delta(H)$)

- c) $\chi(G \square K_3) = \max(\chi(G), 3)$

(Hint: Let $f : V(G) \rightarrow [\chi(G)]$ be a colouring of G and let $g : [3] \rightarrow [3]$ be a colouring of K_3 . Let $k = \max\{\chi(G), 3\}$. Then, show that the map $h((v_1, v_2)) = (f(v_1) + g(v_2)) \bmod k$ is a colouring of $G \square K_3$.)

- d) $\chi(G \square H) = \max(\chi(G), \chi(H))$, where H is any graph

[2+3+3+2 = 10 points]